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CPS 474 - Software and language based security
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Lab 02

Task 1

Steps Taken to complete task:

1. Git pull then Copy Lab 2 folder to my class repo
2. Navigate to AspectJ folder and open it in terminal
3. Modify HelloAspect.aj to add in your name and time
 - a. To do this simply import java.time.LocalDateTime
 - b. Then simply call for the time in the print statement like so: LocalDateTime.now()
4. Run the commands in the screenshot below to compile and run the files.

```
[09/10/26]seed@VM:~/.../AspectJ$ ls
HelloAspect.aj  Hello.java
[09/10/26]seed@VM:~/.../AspectJ$ ajc Hello.java HelloAspect.aj
[09/10/26]seed@VM:~/.../AspectJ$ java -cp ".:usr/share/java/aspectjrt.jar" Hello
Hello AOP test from Grant Harvey. The time now is: 2026-09-10T15:28:22.872229
Hello World!
[09/10/26]seed@VM:~/.../AspectJ$
```

Task 2

Updated the code from Task 1 to add in a function that runs after the other function. You can see how I did this by looking at the code screenshot below

```
[09/10/26]seed@VM:~/.../AspectJ$ ajc Hello.java HelloAspect.aj
[09/10/26]seed@VM:~/.../AspectJ$ java -cp ".:usr/share/java/aspectjrt.jar" Hello
Hello AOP test from Grant Harvey. The time now is: 2026-09-10T15:37:52.842650
Hello World!
This is after Hello.greeting(..) is invoked. The time now is: 2026-09-10T15:37:52.845170
[09/10/26]seed@VM:~/.../AspectJ$
```

```
1 import java.time.LocalDateTime;
2
3 public aspect HelloAspect {
4     pointcut greeting(): call(* Hello.greeting(..));
5
6     before(): greeting() {
7         System.out.print("Hello AOP test from Grant Harvey.
8         The time now is: " + LocalDateTime.now() + " \n");
9     }
10
11    after(): greeting() {
12        System.out.print("This is after Hello.greeting(..) is
13        invoked. The time now is: " + LocalDateTime.now() + "
14        \n");
15    }
16 }
```

Task 3

a: packing java byte code into a jar file

```
[09/10/26] seed@VM:~/.../AspectJ$ javac Hello.java
[09/10/26] seed@VM:~/.../AspectJ$ java Hello
Hello World!
[09/10/26] seed@VM:~/.../AspectJ$ jar -cvfe Hello.jar Hello Hello.class
added manifest
adding: Hello.class(in = 495) (out= 323)(deflated 34%)
[09/10/26] seed@VM:~/.../AspectJ$ java -jar Hello.jar
Hello World!
[09/10/26] seed@VM:~/.../AspectJ$ █
```

b: weaving aspects into a jar file

```
[09/10/26] seed@VM:~/.../AspectJ$ ls
Hello-AOP.jar HelloAspect.aj HelloAspect.class Hello.class Hello.jar Hello.java
[09/10/26] seed@VM:~/.../AspectJ$ ajc -cp ".:usr/share/java/aspectjrt.jar" -inpath Hello.jar
r HelloAspect.aj -outjar Hello-AOP.jar
[09/10/26] seed@VM:~/.../AspectJ$ ls
Hello-AOP.jar HelloAspect.aj HelloAspect.class Hello.class Hello.jar Hello.java
[09/10/26] seed@VM:~/.../AspectJ$ java -cp ".:usr/share/java/aspectjrt.jar" -javaagent:/usr
/share/java/aspectjweaver.jar -jar Hello-AOP.jar
Hello AOP test from Grant Harvey. The time now is: 2026-09-10T15:47:23.686882
Hello World!
This is after Hello.greeting(..) is invoked. The time now is: 2026-09-10T15:47:23.688789
[09/10/26] seed@VM:~/.../AspectJ$ █
```

Task 4:

a:

```
[09/15/26]seed@VM:~/.../ShoppingCart$ ajc -cp ".:usr/share/java/aspectjrt.jar" -inpath ShoppingCart.jar ShoppingCartAspect.aj -outjar ShoppingCart-IRM.jar
[09/15/26]seed@VM:~/.../ShoppingCart$ java -classpath ".:usr/share/java/aspectjrt.jar:usr/share/java/aspectjtools.jar" -javaagent:usr/share/java/aspectjweaver.jar -jar ShoppingCart-IRM.jar
ShoppingCart program is running on port 8888
Waiting for connections from clients
A new customer is connected!
AOP test from Grant Harvey, this is before Wallet.setBalance.
Policy will be enforced by Grant Harvey in AspectJ.
The time now is: 2026-09-15T21:41:46.809064
The balance of the program now is: $29960
█

[09/15/26]seed@VM:~/.../ShoppingCart$ ajc -cp ".:usr/share/java/aspectjrt.jar" -inpath ShoppingCart.jar ShoppingCartAspect.aj -outjar ShoppingCart-IRM.jar
[09/15/26]seed@VM:~/.../ShoppingCart$ java -classpath ".:usr/share/java/aspectjrt.jar:usr/share/java/aspectjtools.jar" -javaagent:usr/share/java/aspectjweaver.jar -jar ShoppingCart-IRM.jar
ShoppingCart program is running on port 8888
Waiting for connections from clients
A new customer is connected!
AOP test from Grant Harvey, this is before Wallet.setBalance.
Policy will be enforced by Grant Harvey in AspectJ.
The time now is: 2026-09-15T21:41:46.809064
The balance of the program now is: $29960
█

[09/15/26]seed@VM:~/.../ShoppingCart$ telnet localhost 8888
Trying 127.0.0.1...
Connected to localhost.
Escape character is '^'.
Welcome to SS-LBS's ShoppingCart. The time now is Tue Sep 15 21:40:47 EDT 2026
Your balance: 30000 credits
Please select your product:
candies - 1
car - 30000
pen - 40
book - 100
What do you want to buy, type e.g., pen?
pen
You have successfully purchased a 'pen'
Your new balance: 29960 credits
Connection closed by foreign host.
[09/15/26]seed@VM:~/.../ShoppingCart$ █
```

b:

I copied the HelloAspect.aj script into a new script, ShoppingCartAspect.aj, and modified it to target Wallet.setBalance(), then recompiled the scripts and ran the ShoppingCart program. I connected to the server and purchased a product, which triggered the before and after advice. The server terminal displayed the AOP messages and updated balance, while the telnet terminal displayed the purchase.

c:

```
import java.time.LocalDateTime;
```

```
public aspect ShoppingCartAspect {
    pointcut getBalance(): call(* Wallet.setBalance(int));

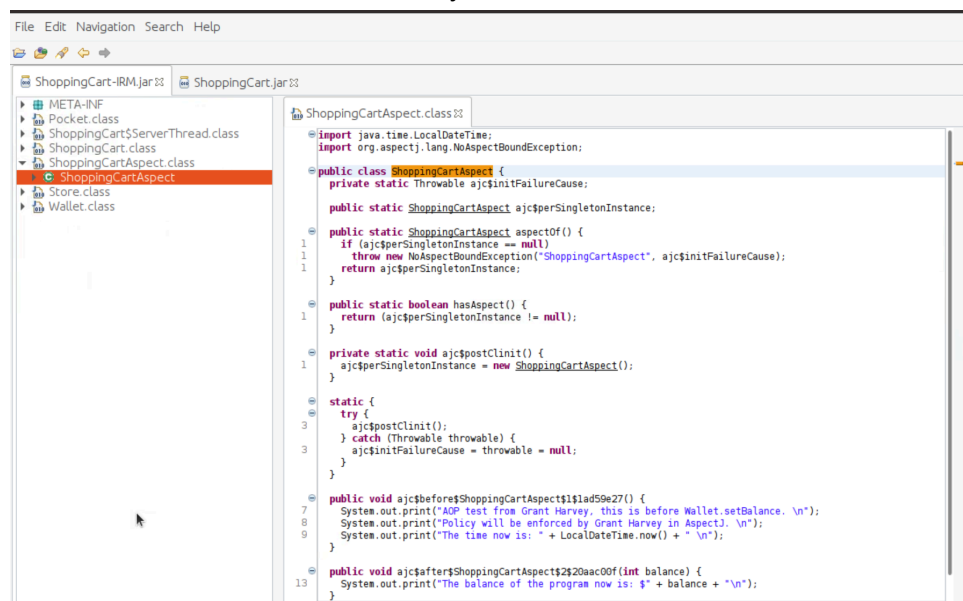
    before(): getBalance() {
        System.out.print("AOP test from Grant Harvey, this is before Wallet.setBalance. \n");
        System.out.print("Policy will be enforced by Grant Harvey in AspectJ. \n");
        System.out.print("The time now is: " + LocalDateTime.now() + " \n");
    }

    after(int balance): call(* Wallet.setBalance(int)) && args(balance) {
        System.out.print("The balance of the program now is: $" + balance + "\n");
    }
}
```

Task 5

a: Aspect Code Examination

ShoppingCartAspect.class is the new class added to ShoppingCart-IRM.jar. Its purpose is to act as the counterpart of the ShoppingCartAspect.aj file, and is used to manage aspect instances and execute advice methods via bytecode hooks.



b: Advice Code

New Advice Code

- Before Advice: Prints the AOP test string, policy enforcement message, and current local date/time.
- After Advice: Prints the updated program balance using the int balance passed to it.

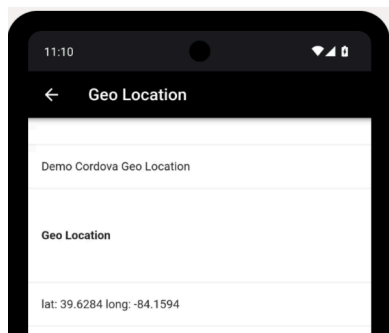
Injection Location: injected directly into "ShoppingCart\$ServerThread.class" surrounding the invocation call site of Wallet.setBalance(int).

c:

Aspect-Oriented Programming (AOP) separates security policies from main code. An Inline Reference Monitor (IRM) uses AOP to rewrite bytecode. It weaves checks directly around operations like modifying a wallet balance, ensuring security rules run automatically and cannot be bypassed.

Task 6

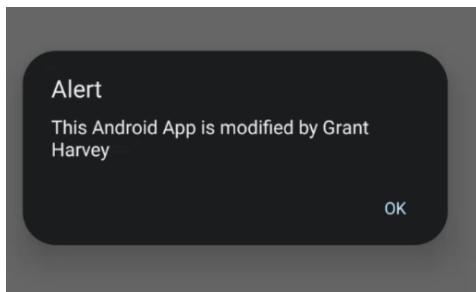
a:



b:

```
6221
6222 tle\", [_vm._v(\"Demo App - Modified By Grant Harvey\")]),\n6223
10865 /*****/ });
10866 alert('This Android App is modified by Grant Harvey');|
```

c: Reverse Engineering mobile apk



Decompiling APKs shows that client-side Android code and resources are fully visible to anyone with basic reverse engineering tools. Because attackers can easily modify app logic and re-sign APKs with their own keys, developers must use code obfuscation and integrity checks rather than relying on secret code.